

## **FMUSER FUQ-4109Y HD IRD Multi-Source Input 1RF DVB-C-DVB-S-S2-ISDB-T, 1ASI 1IP Input MPEG-2 H.264 Decoding ISDB-T Multiplexing CI Descrambling SNMP-Web Management for Broadcast Systems**

Product overview



The FUQ-4109Y Series HD IRD decodes TS signals from ASI, RF tuner (supporting DVB-C/DVB-S/S2/ISDB-T), or IP sources into audio/video. It processes signals through HD decoding compliant with MPEG-2 and H.264 standards, and performs ISDB multiplexing for ASI and IP outputs. The device includes a CI descramble module and supports management via SNMP/web, LCD, and keyboard control. Ideal for broadcast systems requiring reliable signal decoding and multiplexing in headend or distribution environments.

### **Key Feature**

Supports multi-source inputs including RF (DVB-C/DVB-S/S2/ISDB-T), ASI, and IP

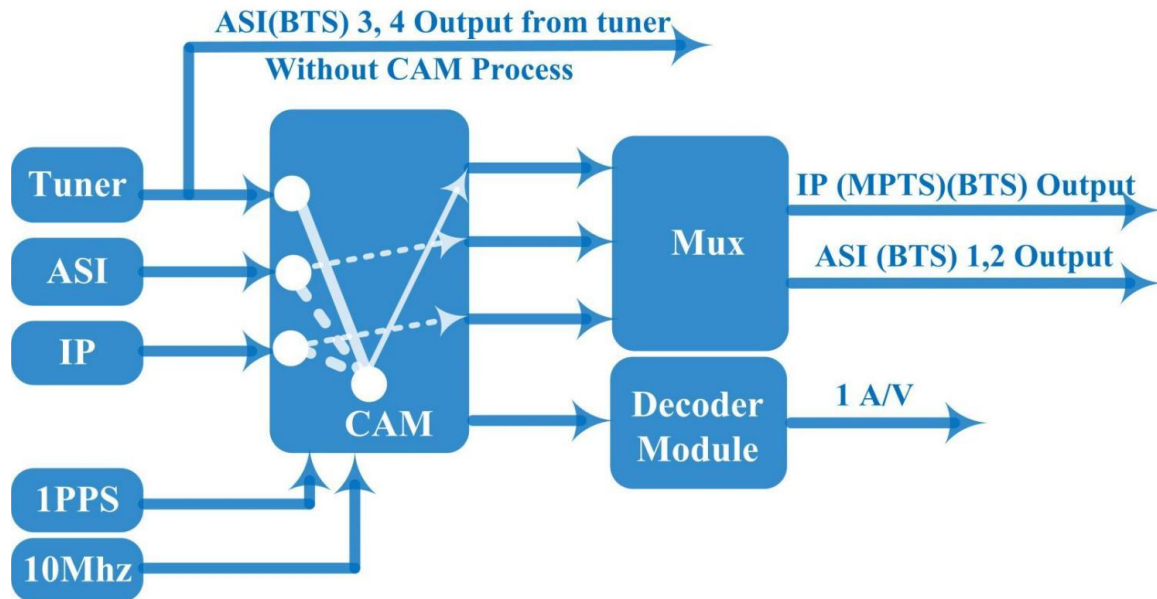
Delivers high-definition decoding up to 1080i with MPEG-2 and H.264 compliance

Performs ISDB-T multiplexing for ASI and IP outputs

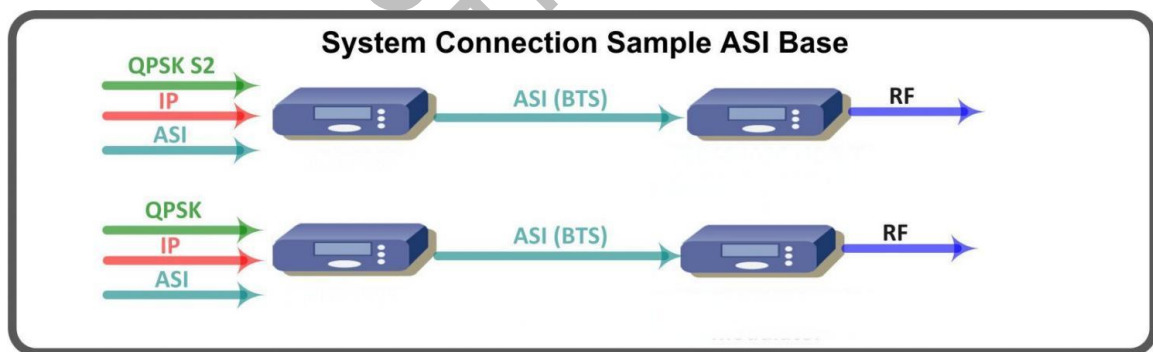
Features CI descrambling with DVB-CSA module via PCMCIA interface

Offers integrated management via SNMP/web, LCD, and keyboard control

Principle Chart



\* The descramled program will be multiplexed



#### Specifications

|       |                                                            |                                                            |                                                            |                                                            |
|-------|------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|
| Input | 1RF Input (DVB-C/DVB-S/S2/ISD B-T optional)                | 1RF Input (DVB-C/DVB-S/S2/ISD B-T optional)                | 1RF Input (DVB-C/DVB-S/S2/ISD B-T optional)                | 1RF Input (DVB-C/DVB-S/S2/ISD B-T optional)                |
| Input | 1ASI Input (support 188/204 package format)                | 1ASI Input (support 188/204 package format)                | 1ASI Input (support 188/204 package format)                | 1ASI Input (support 188/204 package format)                |
| Input | 1 IP MPTS Input (100Mbps, RJ45 Port) over UDP/RTP protocol | 1 IP MPTS Input (100Mbps, RJ45 Port) over UDP/RTP protocol | 1 IP MPTS Input (100Mbps, RJ45 Port) over UDP/RTP protocol | 1 IP MPTS Input (100Mbps, RJ45 Port) over UDP/RTP protocol |
| Input | GPS 10Mhz Clock Input                                      | GPS 10Mhz Clock Input                                      | GPS 10Mhz Clock Input                                      | GPS 10Mhz Clock Input                                      |

| Input                   | GPS 1PPS<br>Signal Input                                                          | GPS 1PPS<br>Signal Input                                                          | GPS 1PPS<br>Signal Input                                                          | GPS 1PPS<br>Signal Input                                                          |
|-------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Demodulating<br>Section | DVB-C                                                                             | Standard                                                                          | Standard                                                                          | J.83A(DVB-C),<br>J.83B, J.83C                                                     |
| Demodulating<br>Section | DVB-C                                                                             | Input Frequency                                                                   | Input Frequency                                                                   | 30 MHz~1000<br>MHz                                                                |
| Demodulating<br>Section | DVB-C                                                                             | Constellation                                                                     | Constellation                                                                     | 16/32/64/128/256<br>QAM                                                           |
| Demodulating<br>Section | DVB-S                                                                             | Input Frequency                                                                   | Input Frequency                                                                   | 950-2150MHz                                                                       |
| Demodulating<br>Section | DVB-S                                                                             | Symbol rate                                                                       | Symbol rate                                                                       | 1-45MSPS                                                                          |
| Demodulating<br>Section | DVB-S                                                                             | Signal Strength                                                                   | Signal Strength                                                                   | - 65- -25dBm                                                                      |
| Demodulating<br>Section | DVB-S                                                                             | Constellation                                                                     | Constellation                                                                     | 1/2, 2/3, 3/4, 5/6,<br>7/8 QPSK                                                   |
| Demodulating<br>Section | DVB-S2                                                                            | Input Frequency                                                                   | Input Frequency                                                                   | 950-2150MHz                                                                       |
| Demodulating<br>Section | DVB-S2                                                                            | Symbol rate                                                                       | Symbol rate                                                                       | 1~45 MSPS                                                                         |
| Demodulating<br>Section | DVB-S2                                                                            | Code rate                                                                         | Code rate                                                                         | 1/2, 3/5, 2/3, 3/4,<br>4/5, 5/6, 8/9, 9/10                                        |
| Demodulating<br>Section | DVB-S2                                                                            | Constellation                                                                     | Constellation                                                                     | QPSK, 8PSK (16<br>APSK, 32 APSK<br>on request)                                    |
| Demodulating<br>Section | DVB-S2                                                                            | Roll off                                                                          | Roll off                                                                          | 0.20, 0.25, 0.35                                                                  |
| Demodulating<br>Section | ISDB-T                                                                            | Input Frequency                                                                   | Input Frequency                                                                   | 30-1000MHz                                                                        |
| ASI Output              | ASI 1&2 output<br>BTS after<br>multiplexing, ASI<br>3&4 for tuner<br>pass-through | ASI 1&2 output<br>BTS after<br>multiplexing, ASI<br>3&4 for tuner<br>pass-through | ASI 1&2 output<br>BTS after<br>multiplexing, ASI<br>3&4 for tuner<br>pass-through | ASI 1&2 output<br>BTS after<br>multiplexing, ASI<br>3&4 for tuner<br>pass-through |
| IP Output               | 1 IP MPTS<br>output (100M,<br>RJ45 Port) over<br>UDP/RTP<br>protocol              | 1 IP MPTS<br>output (100M,<br>RJ45 Port) over<br>UDP/RTP<br>protocol              | 1 IP MPTS<br>output (100M,<br>RJ45 Port) over<br>UDP/RTP<br>protocol              | 1 IP MPTS<br>output (100M,<br>RJ45 Port) over<br>UDP/RTP<br>protocol              |

|                        |                                                                                                           |                                                                                                           |                                                                                                           |                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Video Output           | 1xCVBS(BNC):<br>576i@25fps,<br>480i@29.97fps                                                              | 1xCVBS(BNC):<br>576i@25fps,<br>480i@29.97fps                                                              | 1xCVBS(BNC):<br>576i@25fps,<br>480i@29.97fps                                                              | 1xCVBS(BNC):<br>576i@25fps,<br>480i@29.97fps                                                              |
| Video Output           | 1xYPbPr(BNC):<br>1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps                                       | 1xYPbPr(BNC):<br>1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps                                       | 1xYPbPr(BNC):<br>1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps                                       | 1xYPbPr(BNC):<br>1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps                                       |
| Video Output           | 1xHD/SD-SDI(B<br>NC): 1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps;<br>576i@25fps;<br>480i@29.97fps | 1xHD/SD-SDI(B<br>NC): 1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps;<br>576i@25fps;<br>480i@29.97fps | 1xHD/SD-SDI(B<br>NC): 1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps;<br>576i@25fps;<br>480i@29.97fps | 1xHD/SD-SDI(B<br>NC): 1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps;<br>576i@25fps;<br>480i@29.97fps |
| Video Output           | 1 x HDMI:<br>1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps                                           | 1 x HDMI:<br>1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps                                           | 1 x HDMI:<br>1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps                                           | 1 x HDMI:<br>1080i@25,<br>29.97, 30fps;<br>720p@50,<br>59.94fps                                           |
| Audio Output           | Stereo<br>unbalanced<br>audio: BNC<br>interface                                                           | Stereo<br>unbalanced<br>audio: BNC<br>interface                                                           | Stereo<br>unbalanced<br>audio: BNC<br>interface                                                           | Stereo<br>unbalanced<br>audio: BNC<br>interface                                                           |
| Audio Output           | Stereo balanced<br>audio: XLR<br>interface                                                                | Stereo balanced<br>audio: XLR<br>interface                                                                | Stereo balanced<br>audio: XLR<br>interface                                                                | Stereo balanced<br>audio: XLR<br>interface                                                                |
| Audio Output           | Digital audio:<br>SPDIF                                                                                   | Digital audio:<br>SPDIF                                                                                   | Digital audio:<br>SPDIF                                                                                   | Digital audio:<br>SPDIF                                                                                   |
| Video Decoding         | MPEG-2, H.264                                                                                             | MPEG-2, H.264                                                                                             | MPEG-2, H.264                                                                                             | MPEG-2, H.264                                                                                             |
| Audio Decoding         | MPEG1 layer 2,<br>AC3, HE-AAC<br>(v2), LC-AAC                                                             | MPEG1 layer 2,<br>AC3, HE-AAC<br>(v2), LC-AAC                                                             | MPEG1 layer 2,<br>AC3, HE-AAC<br>(v2), LC-AAC                                                             | MPEG1 layer 2,<br>AC3, HE-AAC<br>(v2), LC-AAC                                                             |
| ISDB-T<br>multiplexing | Support 256 PID<br>mapping per<br>channel<br>(Automatically or<br>manually)                               | Support 256 PID<br>mapping per<br>channel<br>(Automatically or<br>manually)                               | Support 256 PID<br>mapping per<br>channel<br>(Automatically or<br>manually)                               | Support 256 PID<br>mapping per<br>channel<br>(Automatically or<br>manually)                               |
| ISDB-T<br>multiplexing | PCR accurate<br>adjusting                                                                                 | PCR accurate<br>adjusting                                                                                 | PCR accurate<br>adjusting                                                                                 | PCR accurate<br>adjusting                                                                                 |

|                       |                                                     |                                                     |                                                     |                                                     |
|-----------------------|-----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| ISDB-T multiplexing   | Generate PSI/SI table automatically                 | Generate PSI/SI table automatically                 | Generate PSI/SI table automatically                 | Generate PSI/SI table automatically                 |
| ISDB-T BTS parameters | Modes: mode1(2k), mode2(4k), mode3(8k)              | Modes: mode1(2k), mode2(4k), mode3(8k)              | Modes: mode1(2k), mode2(4k), mode3(8k)              | Modes: mode1(2k), mode2(4k), mode3(8k)              |
| ISDB-T BTS parameters | Guard Interval: 1/4, 1/8, 1/16, 1/32                | Guard Interval: 1/4, 1/8, 1/16, 1/32                | Guard Interval: 1/4, 1/8, 1/16, 1/32                | Guard Interval: 1/4, 1/8, 1/16, 1/32                |
| ISDB-T BTS parameters | Code Rate: 1/2, 2/3, 3/4, 5/6, 7/8                  | Code Rate: 1/2, 2/3, 3/4, 5/6, 7/8                  | Code Rate: 1/2, 2/3, 3/4, 5/6, 7/8                  | Code Rate: 1/2, 2/3, 3/4, 5/6, 7/8                  |
| ISDB-T BTS parameters | Constellations: DQPSK/QPSK/16QAM/64QAM              | Constellations: DQPSK/QPSK/16QAM/64QAM              | Constellations: DQPSK/QPSK/16QAM/64QAM              | Constellations: DQPSK/QPSK/16QAM/64QAM              |
| ISDB-T BTS parameters | Layer Mode: A, A+B, A+B+C                           | Layer Mode: A, A+B, A+B+C                           | Layer Mode: A, A+B, A+B+C                           | Layer Mode: A, A+B, A+B+C                           |
| ISDB-T BTS parameters | Number of Segment: 1-13                             | Number of Segment: 1-13                             | Number of Segment: 1-13                             | Number of Segment: 1-13                             |
| ISDB-T BTS parameters | Bandwidth: 6MHz, 7MHz, 8MHz                         | Bandwidth: 6MHz, 7MHz, 8MHz                         | Bandwidth: 6MHz, 7MHz, 8MHz                         | Bandwidth: 6MHz, 7MHz, 8MHz                         |
| SFN parameters        | Maximum time delay: 0ms ~ 1000ms                    | Maximum time delay: 0ms ~ 1000ms                    | Maximum time delay: 0ms ~ 1000ms                    | Maximum time delay: 0ms ~ 1000ms                    |
| SFN parameters        | Device delayed time shifting: -1000ms ~ +1000ms     | Device delayed time shifting: -1000ms ~ +1000ms     | Device delayed time shifting: -1000ms ~ +1000ms     | Device delayed time shifting: -1000ms ~ +1000ms     |
| SFN parameters        | Device ID: 0 ~ 30                                   | Device ID: 0 ~ 30                                   | Device ID: 0 ~ 30                                   | Device ID: 0 ~ 30                                   |
| DVB-CI Parameters     | Descrambler: DVB-CSA                                | Descrambler: DVB-CSA                                | Descrambler: DVB-CSA                                | Descrambler: DVB-CSA                                |
| DVB-CI Parameters     | Smart Card interface: ISO7816                       | Smart Card interface: ISO7816                       | Smart Card interface: ISO7816                       | Smart Card interface: ISO7816                       |
| DVB-CI Parameters     | Interface Card Separation: PCMCIA                   | Interface Card Separation: PCMCIA                   | Interface Card Separation: PCMCIA                   | Interface Card Separation: PCMCIA                   |
| System                | Support SNMP/Web management, LCD + keyboard control | Support SNMP/Web management, LCD + keyboard control | Support SNMP/Web management, LCD + keyboard control | Support SNMP/Web management, LCD + keyboard control |

|         |                                          |                                          |                                          |                                          |
|---------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| System  | English and Chinese interface conversion | English and Chinese interface conversion | English and Chinese interface conversion | English and Chinese interface conversion |
| System  | Ethernet software upgrade                | Ethernet software upgrade                | Ethernet software upgrade                | Ethernet software upgrade                |
| General | Dimensions                               | Dimensions                               | 482mm*455mm*44.5mm                       | 482mm*455mm*44.5mm                       |
| General | Approx weight                            | Approx weight                            | 3.2kg                                    | 3.2kg                                    |
| General | Power                                    | Power                                    | <20W(Max)                                | <20W(Max)                                |
| General | Temperature                              | Temperature                              | 0~45■(Operating), -20~80■(Storage)       | 0~45■(Operating), -20~80■(Storage)       |

#### Order Information:

FU-41091 DVB-C HD IRD;

FU-41092 DVB-S HD IRD;

FU-41095 DVB-S2 HD IRD;

FU-41096 ISDB-T HD IRD;

#### Technical Comparison:

|            |          |                                                                                                 |                                                                                |                                                                                                          |
|------------|----------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
|            |          | FU-408X /FU-400X                                                                                | FU-4109X                                                                       | R&S®AVG050 ISDB-T BTS Gateway                                                                            |
| Stream In  | IP input | Over ,UDP, RTP (100 Base-T)                                                                     | Over UDP/RTP (100 Base-T)                                                      | Over IPv4,UDP, RTP (1000 Base-T)                                                                         |
| Stream Out | ASI out  | ASI 1&2 mirrored output BTS after multiplexing; ASI 3&4 for tuner pass-through                  | ASI 1&2 mirrored output BTS after multiplexing; ASI 3&4 for tuner pass-through | ASI 1&2 mirrored output for BTS in an MPEG-2 transport stream with ancillary data                        |
| Stream Out | IP out   | 8 SPTS output over UDP, 100M Base-T (-FU-408X) 32 SPTS output over UDP, 1000M Base-T (-FU-400X) | 1 IP MPTS output over UDP/RTP (1000 Base-T)                                    | 2 × IP streams with individual IP addresses, UDP ports and FEC configuration in line with SMPTE 2022-1/2 |

|                   |                          |                                                                      |                                                                      |                              |
|-------------------|--------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------|
| Decryption        | BISS                     | Yes                                                                  | N/A                                                                  | Yes                          |
| Reference         | 10MHz REF                | N/A                                                                  | Yes                                                                  | Yes                          |
| Reference         | 1PPS                     | N/A                                                                  | Yes                                                                  | N/A                          |
| ISDB-T Parameters | Modulation Constellation | N/A                                                                  | DQPSK/QPSK/16QAM/64QAM                                               | QPSK/16QAM/64QAM             |
| ISDB-T Parameters | PSI/SI                   | N/A                                                                  | PAT, PMT, NIT, SDT, TOT, TDT                                         | PAT, PMT, NIT, SDT, TOT, BIT |
| ISDB-T Parameters | SFN Adapter              | N/A                                                                  | Yes                                                                  | Yes                          |
| Decoding          | Standard                 | Support H.264 and MPEG2 decode and output V/A via various interfaces | Support H.264 and MPEG2 decode and output V/A via various interfaces | N/A                          |
| Decoding          | LCD Monitor              | Yes                                                                  | Yes                                                                  | N/A                          |

## About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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